

Displacement Sensing Resolution of Position-Sensitive Detectors in Atmospheric Turbulence Using Retroreflected Beam

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Abstract—The position sensing resolution of a lateral-effect photodiode (LEP) and a four-quadrant (4Q) photodetector used as the position sensitive detector (PSD) in a sensor which measures the lateral displacement of a corner cube reflector (CCR) by illuminating it and detecting the position of the reflected light spot with the PSD is studied. An LEP is much noisier than a 4Q detector, but in an outdoor environment the sensitivity of the 4Q detector to atmospheric turbulence due to defocusing makes its resolution worse than that of an LEP. Submillimeter resolutions at the target distances of several hundreds of meters are electrically achievable with an LEP. Outdoors, however, its resolution is also bounded by the atmospheric turbulence. In turbulent measurement conditions, standard deviations of better than 1 cm were typically achievable up to the distance of 300 m and about 10 cm up to half a kilometer.

Index Terms—Angle-of-arrival fluctuations, atmospheric turbulence, displacement sensing, position-sensitive detector (PSD), retroreflected beam.

I. INTRODUCTION

ANALOG position-sensitive detectors (PSD's), i.e., lateral-effect photodiodes (LEP's) or four-quadrant (4Q) detectors, are used for position sensing of light spots in various applications. These include, for example, optical shooting simulators operated outdoors. An important measure of their performance is the lateral position sensing resolution. Generally, better performance should be achievable using a 4Q detector due to its inherently lower noise level. In outdoor applications, however, the effect of atmospheric turbulence and background illumination must be taken into account. A continuous PSD such as LEP provides a better performance when exposed to a high level of background illumination, as the active area required by a 4Q detector is at least four times larger than that of the LEP to cover the same measurement field and therefore it collects more background noise [1]. Here, the effect of atmospheric turbulence on the position sensing resolution of the LEP and 4Q detector is studied.

II. POSITION SENSING RESOLUTION

The reflected beam sensor presented in Fig. 1 includes a parallel axis optical transceiver and a corner cube reflector (CCR) attached to the object the lateral displacement which is to be measured. The measurement field is illuminated with a

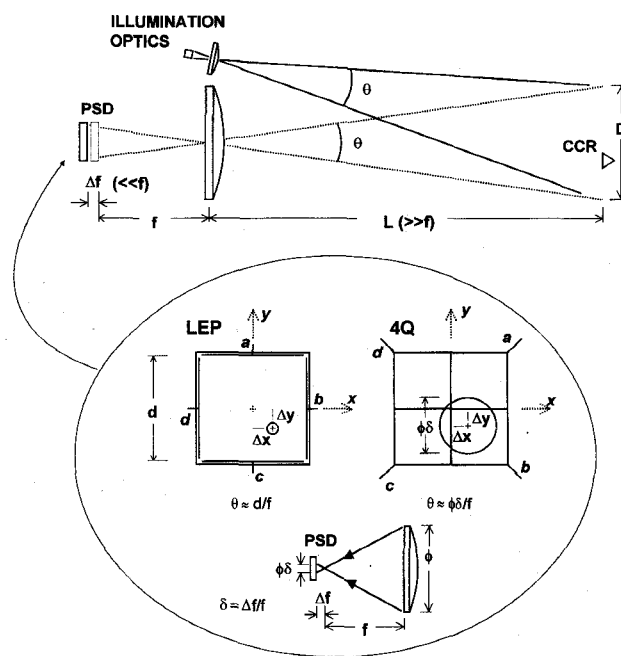


Fig. 1. Operating principles of the reflected beam sensor and the PSD's.

light cone the divergence (θ) of which equals the measurement field of the receiver. The light reflected from the CCR is projected on the PSD located near the focal plane of the receiver. The position of the spot is derived from the signals of the PSD which is either a 4Q detector or an LEP.

The 4Q detector comprises four photodiodes (quadrants) which are positioned symmetrically around the center of the detector and separated by a narrow gap. The position information is derived from the signals received by the quadrants. The measurement field for a 4Q detector is defined by the size of the light spot, as the detector will provide position information only up to the point where the edge of the spot reaches the gap. Therefore, to provide a sufficiently large measurement field, it is essential to enlarge the spot diameter by defocusing the receiver optics.

An LEP is a large-area, single-element photodiode having uniform resistive layers with two wide edge contacts on both the anode and cathode. The current carriers generated in the illuminated region are divided between the electrodes in proportion to the distance of the current paths between the illuminated region and the electrodes. The measurement field of the LEP is determined by the size of its active area,

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and it detects the spot position irrespective of spot size or shape. Therefore, unlike the 4Q detector, the receiver optics are accurately focused in case of the LEP.

If the coordinate system is chosen as shown in Fig. 1, the single-axis displacement of the CCR from the center of the measurement field in case of the 4Q detector and the LEP are

$$\Delta X_{4Q} = \frac{\pi\phi\delta}{8} \frac{L}{f} \frac{(i_a + i_b) - (i_c + i_d)}{i_a + i_b + i_c + i_d}$$

and

$$\Delta X_{LEP} = \frac{d}{2} \frac{L}{f} \frac{i_b - i_d}{i_b + i_d} \quad (1)$$

where i_a , i_b , i_c , and i_d are the average signal currents of the edge contacts (or quadrants) a , b , c , and d ; L is the CCR distance; ϕ is the diameter of the receiver aperture; δ is the relative amount of defocus ($\Delta f/f$); d is the width of the LEP active area; and f is the focal length. Corresponding formulas can be deduced for the perpendicular axis.

A. Electrical Noise

The standard deviation of the lateral displacement at the target distance due to electrical noise is

$$\sigma \approx \frac{kD}{\text{SNR}} \quad (2)$$

where k is a constant describing noise sensitivity of the PSD; $D (= \theta L)$ is the diameter of the measurement field; and SNR the ratio between the root-mean-square (rms) signal and noise currents related to a single edge contact or quadrant, respectively.

The noise sensitivities of the 4Q detector and the LEP are roughly 0.2 and 0.5 [1]. In case of low background illumination, the noise level of an LEP is typically more than ten times higher than that of the 4Q receiver due to the low (typically 10 k Ω) interelectrode resistance. Consequently, in electrical sense the achievable SNR and accordingly the resolution of the 4Q detector is clearly better than that of the LEP.

The electrical resolution can be easily evaluated for the LEP-based sensor since its noise is determined by the high-thermal noise of the interelectrode resistance of the LEP. Thus, it is typically not affected by the preamplifier noise or the shot noise due to background illumination as is the case with the 4Q detector. In terms of system parameters, the single-axis standard deviation of the measured displacement results is simply

$$\sigma \approx \frac{4D^3 \sqrt{4kTB/R_{ie}}}{\tau\phi^2 nPS} \quad (3)$$

where S is the responsivity of the detector; τ is the optical transmittance; P is the rms power of the transmitter; n is the number of CCR's; and R_{ie} is the interelectrode resistance. A high-quality CCR in the near field, a uniform illumination and a fully illuminated receiver lens are assumed. Using typical parameters ($R_{ie} = 10$ k Ω ; $B = 10$ Hz; $\phi = 50$ mm; $n = 1$; $P = 1$ mW; $\tau = 0.5$; and $S = 0.5$ A/W) it can be concluded that submillimeter resolution is achievable if the

reflector distance is within the Fresnel diffraction range and if the width of the measurement field is a few meters. It seems obvious that outdoors such a high resolution may very easily be influenced by atmospheric turbulence.

B. Influence of Atmospheric Turbulence

The randomly fluctuating air temperature produces refractive index inhomogeneities which change the measurement beam intensity distribution and direction as it passes through atmosphere. The spot position on the PSD surface is determined solely by the angle-of-arrival (α) of the received wavefront when a perfectly focused receiver is used [2], and the standard deviation of the measured displacement results is

$$\sigma_{AOA} \approx \sigma_\alpha L \quad (4)$$

where σ_α is the angular standard deviation of the angle-of-arrival fluctuations and a function of turbulence strength described by the refractive index structure coefficient C_n and of several system parameters such as L , ϕ , and θ . A geometric-optics formulation for σ_α in the retroreflected beam case can be found in [3].

In a defocused receiver, additional uncertainty is caused by the illumination fluctuations across the receiver aperture which are directly projected on the light spot on the PSD surface. These spatially noncorrelated intensity fluctuations then cause variation in the light spot centroid. In case of the 4Q detector, the standard deviation of the CCR displacement due to this variation is

$$\sigma_{DF} \approx \frac{\pi\phi\delta}{8} \frac{L}{f} \frac{\sqrt{1-c}}{\text{SFR}} \quad (5)$$

where δ is the relative amount of defocus ($\Delta f/f$); c is the correlation coefficient of the intensity fluctuations between crosswise quadrants of the circular receiver aperture; and SFR is the signal-to-fluctuation ratio related to one quadrant of the receiver aperture defined as the average intensity divided by the rms-value of intensity fluctuations. The correlation depends on the effective distance of the crosswise receiver aperture quadrants in relation to the Fresnel zone size ($\sqrt{L\lambda}$), and the SFR on the path-integrated turbulence strength (C_n, L) and the aperture averaging capability of one quadrant [4].

According to the above, it can be concluded that in a turbulent environment, the resolution of a 4Q detector is impaired more than that of the LEP due to the additional positioning uncertainty caused by defocused receiver optics. Whether and how much this affects the position sensing resolution in a typical working environment obviously depends on the amount of electrical noise and turbulence induced fluctuations present in each case. A set of experiments was performed to examine this.

III. EXPERIMENTAL RESULTS

A. Sensitivity to Atmospheric Turbulence

To compare the turbulence sensitivity of the reflected beam sensors using a 4Q detector and an LEP as a PSD, lateral resolution was measured indoors and outdoors using the same

TABLE I

STANDARD DEVIATIONS OF DISPLACEMENT RESULTS MEASURED USING A STATIONARY CCR TARGET AT 115 m DISTANCE, AND LEP- AND 4Q-BASED REFLECTED BEAM SENSORS WITH EQUAL MEASUREMENT FIELD SIZES

measurement conditions	LEP		4Q	
	horizontal [mm]	vertical [mm]	horizontal [mm]	vertical [mm]
indoors	4.7 - 4.9	4.8 - 5.0	29 - 47	43 - 62
outdoors	8.4 - 8.8	8.2 - 8.7	131 - 146	147 - 169

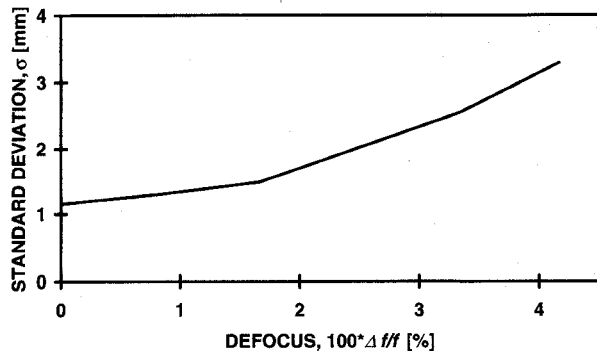


Fig. 2. Standard deviations of atmospheric turbulence limited displacement results as a function of defocus when an LEP is used as PSD.

illumination power and measurement field size in all cases. The noise equivalent powers (NEP) of the 4Q and LEP sensors, including the preamplifier, were about 0.5 pW/√Hz and 20 pW/√Hz, and the distance of the CCR was 115 m. The indoor measurements were performed in a narrow corridor to keep the environment as turbulence-free as possible. The outdoor measurements were made on a sunny Summer day at a temperature of about +20 °C and with a moderate wind.

The results presented in Table I indicate that the 4Q-detector-based sensor is more sensitive to atmospheric turbulence than the LEP sensor and that a better resolution is achieved using an LEP even indoors despite the low level of atmospheric turbulence. The LEP has roughly ten times better resolution indoors, whereas outdoors, where atmospheric turbulence was clearly present, it was about 15 times better. The resolution of the 4Q sensor is upperbounded in both cases by the fluctuations of atmospheric turbulence, while the main noise sources of the LEP sensor were electrical noise indoors and turbulence effects in outdoor experiments. Outdoors, the standard deviations of the 4Q sensor varied from 5% to 7% of the total measurement field, which is clearly larger than the inherently large nonlinearity error of its transfer characteristics, for example.

In another experiment, the turbulence-limited standard deviation of the LEP sensor as a function of receiver defocus was measured. The result presented in Fig. 2 indicates, as expected, that the resolution gets worse as defocusing is increased and that the turbulence sensitivity of the 4Q sensor is not inherent to the detector but is caused by defocusing.

B. Resolution of an LEP Sensor

An additional set of experiments was performed to obtain insight into the effect of atmospheric turbulence on the mea-

TABLE II

STANDARD DEVIATIONS OF DISPLACEMENT RESULTS UNDER DIFFERENT SETS OF OUTDOOR WEATHER CONDITIONS USING A STATIONARY CCR TARGET AT 150 m DISTANCE AND AN LEP SENSOR

weather conditions	horizontal [mm]	vertical [mm]
cloudy morning, +5 °C, moderate wind	2.6 - 3.0	2.5 - 2.9
sunny afternoon, +22 °C, moderate wind	3.0 - 3.7	2.9 - 4.0
cloudy afternoon, +15 °C, moderate wind	2.1 - 2.7	2.2 - 3.0
evening sunset, +20 °C, no wind	0.8 - 1.1	0.8 - 1.2

surement resolution of an LEP sensor which, according to the above results, provides a better resolution than the 4Q detector in a turbulent environment. The optical system consists of a parallel axis transceiver and one CCR. The divergence and the average transmitting power of the illumination cone were about 13 mrad and 1.2 mW. Relatively high-quality receiver optics ($f = 120$ mm, $f/\# = 3$) capable of producing a focal spot of 30 μm in diameter was used in order that the reflected radiation could be focused properly. The CCR had an accuracy of ±2 arcsec and a diameter of 50 mm. The sensor bandwidth was about 30 Hz and the illumination beam was on-off modulated with a frequency of 5 kHz to avoid the contribution of the $1/f$ -noise of the preamplifiers. The dominating noise source of the constructed receiver was the thermal noise of the interelectrode resistance of the LEP and the NEP, accordingly, is about 4 pW/√Hz.

To estimate the contribution of the electrical noise to the sensor resolution, measurements were first performed indoors where atmospheric turbulence was weak. The CCR distance was 150 m and the diameter of the measurement field about 2.5 m. The measured standard deviations varied between 0.68 mm and 0.79 mm. In order to eliminate the effect of atmospheric turbulence, the resolution was first measured using a low enough SNR so that the measurement uncertainty was clearly dominated by electrical noise rather than turbulence induced fluctuations. Thereafter, the SNR was increased to the highest achievable level by using maximum transmitting power (1.2 mW). The contributions of electrical noise and atmospheric turbulence were then calculated taking into account the increased SNR and assuming that the effects were statistically independent. In this measurement, electrical noise (0.5 mm) and atmospheric turbulence (0.5–0.6 mm) had approximately equal contributions. The measured standard deviation due to electrical noise agrees satisfactorily with (3).

The same measurement distance was then used outdoors under varying weather conditions. The beam path was about 1 m above the ground made up of grassland and paved walkway. The results, presented in Table II, show that the measurement resolution is limited by atmospheric turbulence since the measured standard deviations (0.8–4.0 mm) are larger than that caused by electrical noise (0.5 mm) and since they clearly depend on wind and temperature conditions.

The resolution was also measured as a function of distance on a sunny day (+20 °C) with moderate wind. This time the

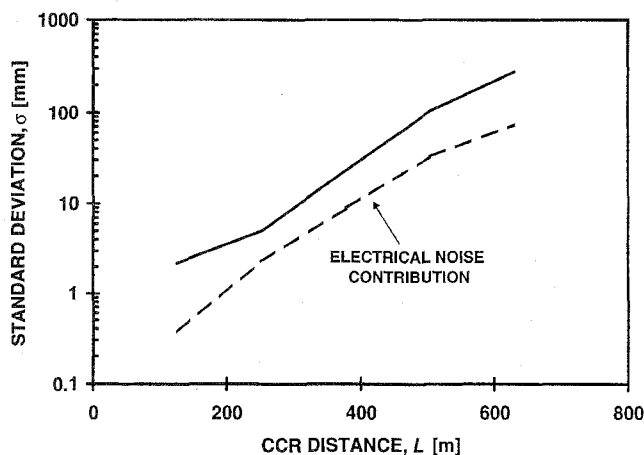


Fig. 3. Position sensing resolution of an LEP sensor as a function of the CCR distance.

beam path was about 1.5 m above a paved walkway. The results, which also include the electrical noise contribution, are presented in Fig. 3. It can be seen that the atmospheric turbulence limits the achievable resolution at all distances and that a resolution better than 1 cm is achievable up to the distance of about 300 m, and the resolution of about 10 cm seems to be achievable at half a kilometer. More elaborate experiments concerning the LEP based sensor can be found in [5].

The effect of atmospheric turbulence was estimated to be caused solely by angle-of-arrival fluctuations of the received beam despite the fact that some degree of defocus was certainly present due to the nonideal performance of the receiver optics. This conclusion was drawn from the fact that the receiver aperture could be half covered without causing in the average displacement results a change no larger than a 1.3 mm at the distance of 150 m, and 4 mm at 500 m, respectively, suggesting that the spatially noncorrelated intensity fluctuations across the receiver aperture could not have any significant effect on the measured standard deviations.

IV. CONCLUSION

The position sensing resolution of an LEP and a 4Q detector used as PSD's in a reflected beam sensor were studied. Due to its low interelectrode resistance, an LEP is much noisier than a 4Q detector and therefore the 4Q detector provides better resolution in an electrical sense than the LEP. An LEP provides, however, far better accuracy in a typical outdoor environment because atmospheric turbulence induced, spatially noncorrelated intensity fluctuations within the light spot which result from defocusing, derange the measurement resolution of the 4Q detector.

The effect of atmospheric turbulence on the measurement resolution of a focused system using an LEP as the PSD was also studied. The results show that, electrically, a submillimeter resolution can easily be achieved using a practically sized measurement field and sensor bandwidth, off-the-shelf components and modest receiver parameters, and that the strength of the atmospheric turbulence determines the achiev-

able resolution outdoors. The measured turbulence-limited standard deviations varied from 0.8–4.0 mm, depending on the temperature and wind when a measurement distance of 150 m was used. Better than 1 cm resolution was achieved at distances up to 300 m and about 10 cm resolution was still achievable at half a kilometer.

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